Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Security+ Certification (SY0-701)

TGS Ref No: TGS-2023039181

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 July 2026	Initial release — Security+ SY0-701 3-day lesson plan aligned to the 36 labs.	Dr. Alfred Ang
2.0	18 August 2026	Reorganised to 5 days aligned to the SY0-701 exam domains (one domain per day); added full concept coverage per exam objective and 7 new labs (43 total); added exam-registration and practice-exam sections.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Security+ Certification (SY0-701)
WSQ Course Reference	TGS-2023039181
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the free Killercoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply general security concepts — control categories and types, the CIA triad and AAA, Zero Trust, change management, cryptographic primitives (hashing, symmetric/asymmetric encryption) and PKI.
- LO2: Identify threats, vulnerabilities and mitigations — threat actors, social engineering, password and web attacks (SQLi, XSS), malware, denial-of-service, wireless attacks and hardening.
- LO3: Implement secure architecture — cloud and enterprise models, network segmentation, firewalls, WAF, VPN, container security, data-at-rest encryption, TLS hardening and resilient backup/RAID.
- LO4: Run security operations — SSH/identity hardening, MFA, SIEM, IDS, EDR, file-integrity monitoring, packet analysis, vulnerability management, incident response, forensics and automation.
- LO5: Manage the security program — governance, risk registers and CVSS, third-party risk, reconnaissance, penetration testing, compliance auditing and security-awareness testing.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 6 questions, 1 hour, open book.
- Practical Performance (PP) — 5 hands-on security tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Domain 1 — General Security Concepts (12%)

Time	Duration	Topic / Activity

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9:30–10:00	30 min	Welcome, course introduction, exam-domain overview, ground rules and mandatory digital attendance (AM)
10:00–11:00	60 min	**Domain 1 concepts: security control categories & types, defence in depth, gap analysis (Obj 1.1)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	**Domain 1 concepts: CIA/DAD triad, AAA, MFA, Zero Trust, physical security, deception; change management (Obj 1.2 –1.3)**
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	**Domain 1 concepts: cryptography — symmetric/asymmetric, hashing, digital signatures, PKI, key protection, obfuscation (Obj 1.4)**
15:00–16:15	75 min	Hands-on: Lab 1: Security Controls Categorization; Lab 2: CIA Triad and AAA on Linux; Lab 3: Hashing, Salting, and Digital Signatures
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 4: Symmetric vs. Asymmetric Encryption; Lab 5: PKI: Build Your Own Certificate Authority
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Domain 2 — Threats, Vulnerabilities & Mitigations (22%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	**Domain 2 concepts: threat actors & motivations, vectors & attack surface; malware, DoS/DDoS, network & application attacks (Obj 2.1–2.2)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	**Domain 2 concepts: social engineering, IOCs, password & wireless attacks, vulnerability types & mitigations (Obj 2.3–2.5). Hands-on: Lab 6: Phishing Email Header Analysis (SPF / DKIM / DMARC); Lab 7: Password Attacks: Spraying, Brute Force, Cracking**
13:00–14:00	60 min	Lunch break

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14:00–16:15	135 min	Hands-on: Lab 8: Malware Static Analysis (Safe Sample); Lab 9: SQL Injection (SQLi); Lab 10: Cross-Site Scripting (XSS)
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 11: DoS Simulation and Detection; Lab 12: Vulnerability Scanning with Nikto and Nuclei; Lab 13: Linux Hardening Baseline with Lynis
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Domain 3 — Security Architecture (18%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	**Domain 3 concepts: architecture models, cloud shared responsibility, IoT/OT; enterprise infrastructure, segmentation, firewalls, WAF, IDS/IPS, VPN (Obj 3.1–3.2)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	**Domain 3 concepts: data protection & states, encryption, DLP; resilience, RAID, backup 3-2-1 (Obj 3.3–3.4). Hands-on: Lab 14: Network Segmentation with Linux Namespaces; Lab 15: Stateful Firewall with UFW and nftables**
13:00–14:00	60 min	Lunch break
14:00–16:15	135 min	Hands-on: Lab 16: Web Application Firewall with ModSecurity; Lab 17: Site-to-Site VPN with WireGuard; Lab 18: Data at Rest: LUKS Full-Disk Encryption; Lab 19: TLS Hardening and Cipher Audit
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 20: Backup and Recovery with restic; Lab 37: Cloud Shared Responsibility & IAM Least Privilege; Lab 38: Container Security with Docker; Lab 39: Storage Resilience with Software RAID (mdadm)
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

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Day 4 — Domain 4 — Security Operations (28%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	**Domain 4 concepts: secure baselines & hardening, asset management; monitoring stack — SIEM, IDS/IPS, EDR, FIM, packet analysis (Obj 4.1–4.4)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 21: SSH Hardening + Fail2ban; Lab 22: IAM with PAM and sudo; Lab 23: Multi-Factor Authentication with TOTP; Lab 24: SIEM Basics with Wazuh
13:00–14:00	60 min	Lunch break
14:00–14:45	45 min	**Domain 4 concepts: vulnerability management & CVSS, IAM/MFA & authorization models, automation/SOAR, incident response & forensics (Obj 4.3–4.8)**
14:45–16:15	90 min	Hands-on: Lab 25: Network IDS with Suricata; Lab 26: Endpoint Detection with osquery; Lab 27: File Integrity Monitoring with AIDE; Lab 28: Vulnerability Management with Trivy
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 29: Incident Response and Forensics; Lab 30: Automation and Orchestration (Mini-SOAR); Lab 40: Packet Capture & Traffic Analysis; Lab 41: Mobile & Endpoint Hardening Baseline; Lab 42: Secure Data Sanitisation (Media Disposal); Lab 43: Digital Forensics — Evidence Acquisition & Hashing
18:15–18:30	15 min	Day 4 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 5 — Domain 5 — Program Management, Oversight & Assessment (20%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	**Domain 5 concepts: governance, policies/standards/procedures; risk management & CVSS,**

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		third-party risk (Obj 5.1–5.3). Hands-on: Lab 31: Risk Register, CVSS Scoring, and Quantitative Risk
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	**Domain 5 concepts: audits & assessments, penetration testing, reconnaissance, security awareness & compliance (Obj 5.4–5.6). Hands-on: Lab 32: Passive Reconnaissance / OSINT; Lab 33: Active Reconnaissance with Nmap**
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Hands-on: Lab 34: Penetration Testing with Metasploit (Authorised Lab); Lab 35: Compliance Audit with OpenSCAP; Lab 36: Security Awareness: Run a Phishing Simulation with GoPhish
15:45–16:00	15 min	Tea break
16:00–16:30	30 min	**Briefing for Assessment and exam-registration walkthrough**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 6 questions, 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — 5 hands-on security tasks, 1 hour, open book. PM digital attendance**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to exam domains)

Domain / exam area	Weighting	Labs
Domain 01: General Security Concepts	12%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5
Domain 02: Threats, Vulnerabilities & Mitigations	22%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
Domain 03: Security Architecture	18%	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 37, Lab 38, Lab 39
Domain 04: Security Operations	28%	Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28, Lab 29, Lab 30, Lab 40, Lab 41, Lab 42, Lab 43
Domain 05: Security Program Management & Oversight	20%	Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36

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